

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250283831

Kind Code

A1

Publication Date

September 11, 2025

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SYSTEM AND METHOD FOR DETECTING CONTAMINATION OF THIN-FILMS

Abstract

A thin-film deposition system deposits a thin-film on a wafer. A radiation source irradiates the wafer with excitation light. An emissions sensor detects an emission spectrum from the wafer responsive to the excitation light. A machine learning based analysis model analyzes the spectrum and detects contamination of the thin-film based on the spectrum.

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Appl. No.: 19/217595

Filed: May 23, 2025

Related U.S. Application Data

parent US continuation 18324021 20230525 parent-grant-document US 12320755 child US 19217595

parent US continuation 17169101 20210205 parent-grant-document US 11686683 child US 18324021

us-provisional-application US 63018312 20200430

Publication Classification

Int. Cl.: G01N21/84 (20060101); B82Y30/00 (20110101); C23C14/54 (20060101); G01N21/64 (20060101); G01N21/95 (20060101); G06F30/27 (20200101); G06N3/08 (20230101); G06V10/145 (20220101)

U.S. Cl.:

CPC **G01N21/8422** (20130101); **C23C14/548** (20130101); **G01N21/6489** (20130101);
G01N21/9501 (20130101); **G06F30/27** (20200101); **G06N3/08** (20130101);
G06V10/145 (20220101); B82Y30/00 (20130101); G01N2021/6491 (20130101)

Background/Summary

BACKGROUND

Technical Field

[0001] The present disclosure relates to the field of thin-film deposition.

Description of the Related Art

[0002] There has been a continuous demand for increasing computing power in electronic devices including smart phones, tablets, desktop computers, laptop computers and many other kinds of electronic devices. Integrated circuits provide the computing power for these electronic devices. One way to increase computing power in integrated circuits is to increase the number of transistors and other integrated circuit features that can be included for a given area of semiconductor substrate.

[0003] To continue decreasing the size of features in integrated circuits, various thin-film deposition techniques are implemented. These techniques can form very thin-films. However, thin-film deposition techniques also face serious difficulties in ensuring that the thin-films are properly formed.

Description

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

[0004] FIG. 1 is a block diagram of a thin-film deposition system, according to one embodiment.

[0005] FIG. 2 is an illustration of a thin-film deposition system, according to one embodiment.

[0006] FIG. 3 is an illustration of a thin-film deposition system, according to one embodiment.

[0007] FIG. 4 is a cross-sectional view of a semiconductor wafer, according to one embodiment.

[0008] FIG. 5 is a cross-sectional view of a semiconductor wafer, according to one embodiment.

[0009] FIG. 6 is a block diagram of a control system, according to one embodiment.

[0010] FIGS. 7-13 are flow diagrams of methods for detecting defects in thin-films, according to various embodiments

DETAILED DESCRIPTION

[0011] In the following description, many thicknesses and materials are described for various layers and structures within an integrated circuit die. Specific dimensions and materials are given by way of example for various embodiments. Those of skill in the art will recognize, in light of the present disclosure, that other dimensions and materials can be used in many cases without departing from the scope of the present disclosure.

[0012] The following disclosure provides many different embodiments, or examples, for implementing different features of the described subject matter. Specific examples of components and arrangements are described below to simplify the present description. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat

reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

[0013] Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

[0014] In the following description, certain specific details are set forth in order to provide a thorough understanding of various embodiments of the disclosure. However, one skilled in the art will understand that the disclosure may be practiced without these specific details. In other instances, well-known structures associated with electronic components and fabrication techniques have not been described in detail to avoid unnecessarily obscuring the descriptions of the embodiments of the present disclosure.

[0015] Unless the context requires otherwise, throughout the specification and claims that follow, the word “comprise” and variations thereof, such as “comprises” and “comprising,” are to be construed in an open, inclusive sense, that is, as “including, but not limited to.”

[0016] The use of ordinals such as first, second and third does not necessarily imply a ranked sense of order, but rather may only distinguish between multiple instances of an act or structure.

[0017] Reference throughout this specification to “one embodiment” or “an embodiment” means that a particular feature, structure or characteristic described in connection with the embodiment is included in at least one embodiment. Thus, the appearances of the phrases “in one embodiment” or “in an embodiment” in various places throughout this specification are not necessarily all referring to the same embodiment. Furthermore, the particular features, structures, or characteristics may be combined in any suitable manner in one or more embodiments.

[0018] As used in this specification and the appended claims, the singular forms “a,” “an,” and “the” include plural referents unless the content clearly dictates otherwise. It should also be noted that the term “or” is generally employed in its sense including “and/or” unless the content clearly dictates otherwise.

[0019] Embodiments of the present disclosure provide thin-films of reliable thickness and composition. Embodiments of the present disclosure utilize machine learning techniques to detect contamination or other defects in thin-films. The contamination can be detected in-situ, thereby enabling immediate cessation of thin-film deposition processes immediately after a defective deposition process. Accordingly, rather than having a large number of wafers receive defective thin-films before testing can detect a problem, problems are detected right away and additional wafers will not be affected. Integrated circuits that include the thin-films will not have performance problems that can result if the thin-films are not properly formed. Furthermore, batches of semiconductor wafers will have improved yields and fewer scrapped wafers.

[0020] FIG. 1 is a block diagram of a thin-film deposition system **100**, according to one embodiment. The thin-film deposition system **100** includes a thin-film deposition chamber **102** defining an interior volume **103**. The thin-film deposition system **100** includes deposition equipment **104** configured to perform a thin-film deposition process on a wafer **106** positioned within the interior volume **103**. The thin-film deposition system **100** includes a radiation source **108**, an emissions sensor **110**, and a control system **112**. The radiation source **108**, the emissions sensor **110**, and the control system **112** cooperate to detect contamination of a thin-film formed on the wafer **106**.

[0021] The thin-film deposition equipment **104** can include deposition equipment positioned outside the thin-film deposition chamber **102**, inside the thin-film deposition chamber **102**, or both

inside and outside the thin-film deposition chamber **102**. The deposition equipment **104** can include equipment for performing one or more of physical vapor deposition (PVD) processes, chemical vapor deposition (CVD) processes, atomic layer deposition (ALD) processes, or other types of thin-film deposition processes for forming a thin-film on the wafer **106**.

[0022] In some cases it is possible that a thin-film may be contaminated during or after a thin-film deposition process. Contamination may occur due to contaminants or impurities in a target material, a precursor material, or other materials introduced into the thin-film deposition chamber **102** during a thin-film deposition process. In many cases, the thin-film deposition processes are performed in vacuum conditions. It is possible that an air leak or another flaw in equipment can allow outside air to enter into the thin-film deposition chamber **102** during a thin-film deposition process in which vacuum conditions are intended. In this case, it is possible that oxygen in the air will oxidize the thin-film. If oxidation of the thin-film occurs, the thin-film may not have the intended structure, composition, and characteristics to perform a selected function in the wafer **106**.

[0023] The thin-film deposition system **100** utilizes the radiation source **108**, the emissions sensor **110**, and the control system **112** to detect contamination of the thin-films. The radiation source **108** outputs an excitation light **109**. The excitation light **109** irradiates the thin-film formed on the wafer **106**. Irradiation of the thin-film formed on the wafer **106** can excite the atoms or molecules of the thin-film. The excited atoms or molecules of the thin-films may output emissions **111** responsive to being energized by the excitation light **109**. The emissions **111** can include light or particles, such as electrons, or a combination of light and particles. The emissions **111** collectively have an energy spectrum. The energy spectrum corresponds to the spectrum of emitted light or the energy spectrum of emitted particles. If the emissions **111** include photons, the energy spectrum corresponds to a photon spectrum including the various wavelengths associated with emission photons. If the emissions **111** include particles, such as electrons, then the energy spectrum corresponds to the energies of the emitted particles. The energy spectrum can provide an indication of the types of atoms, molecules, or compounds included in the thin-film.

[0024] As used herein, the terms “radiation” and “excitation light” refer to electromagnetic radiation inside or outside of the visible spectrum. Accordingly, the excitation light **109** can include electromagnetic radiation outside the visible spectrum.

[0025] Although FIG. **1** illustrates that the radiation source **108** is positioned within the thin-film deposition chamber **102**, the radiation source **108** may be positioned outside of the thin-film deposition chamber **102**. The radiation source **108** may be positioned partially within the thin-film deposition chamber **102** and partially outside of the thin-film deposition chamber **102**. If the radiation source **108** is positioned outside of the thin-film deposition chamber **102**, the radiation source **108** may irradiate the wafer **106** via one or more windows or openings in a wall of the thin-film deposition chamber **102**.

[0026] The emissions sensor **110** is configured to sense the emissions **111**. The thin-film may emit emissions **111** in various directions. The emissions sensor **110** is positioned so that some of the emissions **111** will be received by the emissions sensor **110**. The emissions sensor **110** senses the wavelengths of the various emissions **111** received by the emissions sensor **110**. The emissions sensor **110** outputs sensor signals indicative of the spectrum of the emissions **111**.

[0027] Although FIG. **1** illustrates that the emissions sensor **110** is positioned within the thin-film deposition chamber **102**, the emissions sensor **110** may be positioned outside of the thin-film deposition chamber **102**. The emissions sensor **110** may be positioned partially within the thin-film deposition chamber **102** and partially outside of the thin-film deposition chamber **102**. If the emissions sensor **110** is positioned outside of the thin-film deposition chamber **102**, the emissions sensor **110** may receive emissions **111** via one or more windows or openings in a wall of the thin-film deposition chamber **102**.

[0028] The control system **112** is coupled to the radiation source **108**, the emissions sensor **110**, and the deposition equipment **104**. The control system **112** receives the sensor signals from the

emissions sensor **110**. The control system **112** analyzes the sensor signals and determines the spectrum of the emissions **111**. The control system **112** can detect contamination of the thin-film based on the spectrum of the emissions **111**.

[0029] The control system **112** includes the analysis model **114**. The analysis model **114** is trained with a machine learning process to detect contamination of the thin-film based on the emission spectrum sensed by the emissions sensor **110**. The analysis model **114** can include a neural network or another type of machine learning model. As is set forth in more detail with regards to FIGS. **6** and **7**, the analysis model **114** is trained with a training set that includes a plurality of spectra each associated with contaminated or non-contaminated thin-films. The machine learning process utilizes the training set to train the analysis model **114** to reliably detect contamination of the thin-film based on the emission spectrum sensed by the emissions sensor **110**.

[0030] If the control system **112** detects contamination of the thin-film, the control system **112** can take various responsive actions. For example, the control system **112** can cause the thin-film deposition system **100** to stop operating based on detection of contamination in the thin-film. The control system **112** can output information indicating the type of contamination. For example, if the analysis model **114** detects that the spectrum indicates unwanted oxidation of the thin-film, then the control system **112** can indicate oxygen contamination. If the analysis model **114** detects other types of contamination, then the control system **112** can output information indicating other types of contamination.

[0031] The control system **112** can detect contamination of the thin-film in-situ. In other words, the control system **112** can detect contamination of the thin-film during the thin-film deposition process or shortly after the thin-film deposition process while the wafer **106** is still in the thin-film deposition chamber. This can provide tremendous benefits because each wafer **106** can be monitored and contamination can be detected immediately rather than at a later time when a large number of wafers **106** may have been processed and contaminated. This increases wafer yields and reduces the number of scrapped wafers.

[0032] The control system **112** can include processing, memory, and information transmission resources. The processing, memory, and information transmission resources can be located at a facility of the thin-film deposition system **100**. Alternatively, the processing, memory, and information transmission resources can be located remotely from the facility of the thin-film deposition system **100**. The control system **112** may be a distributed control system including resources in multiple locations. The control system **112** can include cloud-based resources and local physical resources.

[0033] FIG. **2** is a block diagram of a thin-film deposition system **200**, according to one embodiment. The thin-film deposition system **200** is similar to the thin-film deposition system **100** of FIG. **1**, except that the thin-film deposition system **200** is a PVD system implementing an excitation laser and a light sensor. The system **100** can utilize the components, systems, and processes described in relation to FIG. **2**. The thin-film deposition system **200** includes a thin-film deposition chamber **202** defining an interior volume **203**. A wafer support **222** supports a wafer **206** within the interior volume **203** of the thin-film deposition chamber **202**.

[0034] In the example of FIG. **2**, the thin-film deposition system **200** is a PVD sputtering deposition system, though other types of deposition systems can be utilized without departing from the scope of the present disclosure. The thin-film deposition system **200** includes a sputtering magnetron cathode **216** and a sputtering target **218**. During the deposition process, a voltage source **220** applies a voltage signal to the sputtering magnetron cathode **216**. The result is that atoms are ejected from the sputtering target **218**. The wafer support **222** acts as a grounded anode of the sputtering system. Atoms from the sputtering target **218** accumulate on the surface of the wafer **206**. The result is that a thin-film is deposited on the wafer **206**.

[0035] The thin-film deposition system **200** also includes an exhaust channel **224** communicatively coupled to the interior volume **203** of the thin-film deposition chamber **202**. A valve **226** couples

the exhaust channel **224** to a pump **228**. Prior to the thin-film deposition process, the valve **226** is opened and the pump **228** is activated. The pump **228** generates a vacuum within the interior volume **203** by pumping fluids from the interior volume **203**. When the interior volume is below a threshold pressure, i.e., substantially at vacuum, the PVD process can begin.

[0036] In one example, the thin-film deposition process is a titanium nitride deposition process. In this case, the sputtering target **218** is a titanium sputtering target. After the interior volume is evacuated by the pump **228**, nitrogen is flowed into the interior volume **203**. The sputtering process begins in the presence of the nitrogen. The result is that a titanium nitride thin-film is deposited on the wafer **206**.

[0037] In some cases it is possible that due to a leak in the thin-film deposition chamber **202**, air can enter into the interior volume **203** during the deposition process. If this occurs, then the titanium nitride thin-film may oxidize, resulting in the presence of titanium oxide in the thin-film. The presence of titanium oxide in the thin-film will result in the thin-film not being able to perform its function as a barrier layer or glue layer, as the case may be.

[0038] In order to detect oxidation, or other contamination of the thin-film, the thin-film deposition system **200** includes an excitation laser **208** and a light sensor **210**. The excitation laser **208** and the light sensor **210** may be positioned in a tube **213** that extends from an exterior of the thin-film deposition chamber **202** to the interior of the thin-film deposition chamber **202**. The interior surface of the tube **213** may be substantially reflective for most of the extent of the tube **213**. An end of the tube **213** may be transparent such that excitation light and emission photons can pass through. In FIG. 2, the transparent portion of the tube is indicated by dashed lines. In one example, the transparent portion of the tube differs from the reflective portion of the tube in that the transparent portion of the tube **213** does not include a reflective coating, while the reflective portion of the tube **213** includes a reflective coating.

[0039] The excitation laser **208** outputs excitation light **209**. The excitation light reflects within the reflective portion of the tube. At the nonreflective portion of the tube, the excitation light **209** passes from the tube **213** into the interior volume **203** of the thin-film deposition chamber **202**. The excitation light **209** then irradiates the wafer **206**. In particular, the excitation light **209** irradiates the thin-film deposited on the wafer **206**.

[0040] Some of the excitation light **209** is absorbed by the thin-film. The result is that valence electrons in the atoms or compounds of the thin-film transition from lower energy levels to higher energy levels. Subsequently, the electrons return from the higher energy levels to the lower energy levels. When the electrons return from the higher energy levels to the lower energy levels, the electrons emit emission photons **211** with an energy corresponding to the difference in the energy levels.

[0041] The spectrum of emission photons **211** emitted by the thin-film is indicative of the composition of the thin-film. Different materials will emit photons of different wavelengths based on the atoms or molecules that make up the materials. Accordingly, the photon spectrum of the emission photons **211** from the thin-film responsive to irradiation with the excitation light is indicative of the composition of the thin-film.

[0042] The light sensor **210** is configured to receive and sense the emission photons **211**. In particular, the light sensor **210** is positioned at the transparent end of the tube **213** such that emission photons **211** are able to pass through the tube **213** and be received by the light sensor **210**. The light sensor **210** can be configured to sense light in a range of wavelengths corresponding to an expected range of wavelengths for emission photons to **211** from contaminated and non-contaminated thin-films.

[0043] Returning to the example in which the thin-film deposition system **200** deposits a titanium nitride thin-film, the excitation laser **208** emits an excitation light of between 300 nm and 330 nm. This range is selected because one possible contamination of titanium nitride is oxidation of the titanium nitride. Oxidation of the titanium nitride results in titanium oxide, which is a

semiconductor material with a bandgap of about 3.2 eV. Photons in the range of 300 nm and 330 nm have sufficient energy (~3.7 eV-4.1 eV) to excite electrons to in the valence band to transition to the conduction band. Accordingly, the excitation laser **208** is selected so that the excitation photons have an energy greater than the bandgap of titanium oxide. Electrons in the valence band can absorb photons of the excitation light and transition through the bandgap to the conduction band. When the electrons transition back from the conduction band to the valence band, the electrons will emit emission photons **211** with an energy corresponding to the difference between the energy level of the valence band and the energy level from which the electron returned in the conduction band. [0044] In one example, TiO.sub.2 has a photoluminescence spectrum with an intensity peak centered about 360 nm. Accordingly, analysis of the spectrum of emission photons **211** can indicate whether the spectrum corresponds to TiO.sub.2, indicating contamination of the thin-film by oxidation. In this example, the light sensor **210** can be an ultraviolet radiation sensor with particular sensitivity to ultraviolet light in a range between 200 nm and 400 nm. Correspondingly, the excitation laser **208** can be an ultraviolet laser.

[0045] While a particular example has been given of a titanium nitride thin-film contaminated by oxidation, the thin-film deposition system **200** can include many other types of deposition processes, thin-films, contamination detections, radiation sources and radiation sensors that the particular examples described herein, without departing from the scope of the present disclosure. For example, the thin film can include Ti, TiAl, TiON, TiAlO, TiAl, TiAlC, a middle end of line (MEOL) contact metal, a front end of line (FEOL) high-K cap or metal gate, or other types of thin films.

[0046] The control system **212** is coupled to the excitation laser **208**, the light sensor **210**, and the voltage source **220**. The control system **212** receives sensor signals from the light sensor **210**. The sensor signals from the light sensor **210** indicate the spectrum of emission photons **211** received by the light sensor **210** from the thin-film. The control system **212** analyzes the sensor signals to determine whether the spectrum of emission photons corresponds to a contaminated thin-film or properly formed thin-film. If the control system **212** determines that the photon spectrum corresponds to a contaminated thin-film, then the control system **212** can shut down the thin-film deposition system **200** and can output an alert. In the example of oxidation of a titanium nitride thin-film, the alert can indicate a leak in the thin-film deposition chamber **202**.

[0047] The control system **212** includes an analysis model **214**. The analysis model **214** is trained with a machine learning process to reliably detect photon spectra indicating various types of contamination or properly formed thin-films. Further details regarding training of an analysis model are provided in relation to FIGS. 6 and 7.

[0048] While FIG. 2 illustrates an excitation laser **208** and a light sensor **210** that senses emission photons, the thin-film deposition system **200** can include other types of excitation sources and radiation sensors. In one embodiment, the excitation source is an x-ray source that emits x-ray radiation. The x-ray source irradiates the thin-film with x-rays. X-rays are high energy photons. The high-energy x-ray photons energize the thin-film and cause the thin-film to emit electrons via the photoelectric effect.

[0049] In the example of an x-ray source, the light sensor **210** is, instead, a photoelectron sensor. The emissions from the thin-film responsive to the x-rays are photoelectrons. The photoelectron sensor can determine the energy of the electrons received from the thin-film via the photoelectric effect. Because the energy of the x-rays is known and because the energy of the emitted electrons is measured by the photoelectron sensor, the electron binding energy of the emitted electrons can be determined. The electron binding energy of electrons emitted from the thin-film is indicative of the material of the thin-film. Accordingly, by analyzing the spectrum of energies of the emitted electrons, the composition of the thin-film can be determined. In this case, the analysis model can determine whether the thin-film was contaminated based on the energy spectrum of the photoelectrons.

[0050] FIG. 3 is an illustration of a thin-film deposition system **300**, according to one embodiment. The thin-film deposition system **300** is similar in many ways to the thin-film deposition systems **100**, **200** of FIGS. 1 and 2, except that the thin-film deposition system **300** includes separate chambers for thin-film deposition and thin-film measurement. The thin-film deposition system **300** includes a thin-film deposition chamber **302** and a thin-film analysis chamber **334**. The thin-film deposition system **300** includes deposition equipment **304** configured to perform a thin-film deposition process on a wafer **306** supported by a support **322** in the thin-film deposition chamber **302**.

[0051] The thin-film analysis chamber **334** and the thin-film deposition chamber **302** are communicatively coupled by a transfer channel **332**. After a thin-film deposition process has been performed on the wafer **306**, the wafer **306** is transported via the transfer channel **332** to the thin-film analysis chamber **334**. The thin-film analysis chamber **334** includes a support **330**. The wafer **306** is positioned on the support **330** after being transferred from the thin-film deposition chamber **302**. The wafer **306** can be transferred by a robot arm (not pictured in FIG. 3).

[0052] In one embodiment, because the thin-film deposition chamber **302** and the thin-film analysis chamber **334** are coupled together by the transfer channel **332**, the vacuum or pressure condition in the thin-film deposition chamber **302** is communicated to the thin-film analysis chamber **334**. This means that when the wafer **306** is transferred from the thin-film deposition chamber **302** to the thin-film analysis chamber **334**, the wafer **306** does not pass through an additional contamination environment.

[0053] A radiation source **308** and the emissions sensor **310** are positioned in or adjacent to the thin-film analysis chamber **334**. The radiation source **308** is configured to irradiate the wafer **306** with excitation light. The emissions sensor **310** is configured to receive and sense emissions from the wafer **306** responsive to the excitation light. The radiation source **308** and the emissions sensor **310** can include the same types of radiation sources and emissions sensors as those described in relation to FIGS. 1 and 2.

[0054] The control system **312** can receive the emission spectrum signals from the emissions sensor **310**. The analysis model **314** can analyze the spectrum and determine whether or not the thin-film is contaminated. If the thin-film is contaminated, the control system **312** can output an alert and stop operation of the thin-film deposition system **300**.

[0055] FIG. 4 is a cross-sectional view of a wafer **406**, according to one embodiment. The wafer **406** includes a nanosheet structure **440**. An interfacial layer **442** is positioned over the nanosheet structure. A high K gate dielectric layer **444** is positioned on the interfacial layer **442** and on sidewalls of a trench **448** formed in a layer of dielectric material **445**. The trench **448** is formed to be filled by a gate electrode of a nanosheet transistor.

[0056] Before deposition of the gate electrode, a layer of titanium nitride **446** is formed on the high K gate dielectric layer **444**. The layer of titanium nitride acts as a work function layer for the gate electrode that will be formed in the trench **448**. The titanium nitride increases the work function and improves the overall function of the nanosheet transistor.

[0057] The layer of titanium nitride **446** is an example of a thin-film layer formed in the thin-film deposition systems of FIGS. 1-3. After formation of the layer of titanium nitride **446**, a radiation source irradiates the layer of titanium nitride **446** with excitation light **409**. The excitation light **409** can correspond to the various examples of excitation light given in relation to FIGS. 1-3.

Accordingly, the excitation light can include ultraviolet radiation, x-ray radiation, or other types of radiation emitted from a radiation source.

[0058] The layer of titanium nitride **446** absorbs some of the excitation light **409** and outputs emissions **411**. The emissions **411** can include photoluminescence photons emitted from the layer of titanium nitride **446** responsive to absorbing the excitation light **409**. Alternatively, the emissions **411** can include photoelectrons ejected from the layer of titanium nitride **446** responsive to absorbing the excitation light **409**.

[0059] A radiation sensor (not shown) can receive and sense the emissions **411**. The radiation sensor can include the types of emissions sensors described in relation to FIGS. **1-3**, or other types of radiation sensors. The radiation sensor can provide sensor signals to a control system including an analysis model as described in relation to FIGS. **1-3**. The control system and the analysis model can determine whether the titanium nitride has been contaminated based on the spectrum of the emissions **411**, as described previously in relation to FIGS. **1-3**, and as will be described in further detail in relation to FIGS. **6** and **7**.

[0060] FIG. **5** is a cross-sectional view of a wafer **506**, according to one embodiment. The wafer **506** includes a nanosheet structure **540**. An interfacial layer **542** is positioned over the nanosheet structure **540**. A high K dielectric layer **544** is positioned on the interfacial layer **542**. Sidewall spacers **551** are positioned adjacent to the high K dielectric layer **544**. A work function layer of titanium nitride **546** is positioned on the high K dielectric layer **544**. The gate electrode **550** is formed in contact with the work function layer of titanium nitride **546**. Trenches **552** are formed in a layer of dielectric material **555**. The trenches **552** are for source and drain electrodes. A source region **553** at the drain region **557** or adjacent to the nanosheet structure **540**. Silicide layers **554** and **556** are positioned at the source and drain regions by **53** and **557**. A glue layer of titanium nitride **556** is formed on the layer of dielectric material **555**, on the sidewalls of the trenches **552**, and on the gate electrode **550**.

[0061] The layer of titanium nitride **556** is an example of a thin-film layer formed in the thin-film deposition systems of FIGS. **1-3**. After formation of the layer of titanium nitride **556**, a radiation source irradiates the layer of titanium nitride **556** with excitation light **509**. The excitation light **509** can correspond to the various examples of excitation light given in relation to FIGS. **1-3**. Accordingly, the excitation light can include ultraviolet radiation, x-ray radiation, or other types of radiation emitted from a radiation source.

[0062] The layer of titanium nitride **556** absorbs some of the excitation light **509** and outputs emissions **511**. The emissions **511** can include photoluminescence photons emitted from the layer of titanium nitride **556** responsive to absorbing the excitation light **509**. Alternatively, the emissions **511** can include photoelectrons ejected from the layer of titanium nitride **556** responsive to absorbing the excitation light **509**.

[0063] A radiation sensor (not shown) can receive and sense the emissions **511**. The radiation sensor can include the types of emissions sensors described in relation to FIGS. **1-3**, or other types of radiation sensors. The radiation sensor can provide sensor signals to a control system including an analysis model as described in relation to FIGS. **1-3**. The control system and the analysis model can determine whether the titanium nitride has been contaminated based on the spectrum of the emissions **511**, as described previously in relation to FIGS. **1-3**, and as will be described in further detail in relation to FIG. **6**.

[0064] While particular layers of titanium nitride are described in relation to FIGS. **4** and **5**, principles of the present disclosure extend to other types of thin-films deposited and other types of structures and with other types of deposition processes. Detecting contamination with an analysis model, a radiation source, and an emissions sensor can be utilized for a large variety of thin-films and many different types of contamination.

[0065] FIG. **6** is a block diagram of a control system **612**, according to one embodiment. The control systems **112**, **212**, and **312** of FIGS. **1-3** can include the components and functionality of the control system **612**. Accordingly, the control system **612** can be implemented with the systems, processes and components described in relation to FIGS. **1-5**. The control system **612** of FIG. **6** is configured to control operation of a thin-film deposition system, according to one embodiment. The control system **612** utilizes machine learning to determine whether or not a thin-film is contaminated or otherwise improperly formed. The control system **612** can detect contamination of the thin-film and stop further thin-film deposition processes before additional wafers can be affected with in properly formed thin-films. The control system **612** can raise an alert indicating a

problem with the thin-film deposition process.

[0066] In one embodiment, the control system **612** includes an analysis model **614** and a training module **660**. The training module trains the analysis model **614** with a machine learning process. The machine learning process trains the analysis model **614** to detect whether or not a thin-film is contaminated based on emission spectra from the thin-film. Although the training module **660** is shown as being separate from the analysis model **614**, in practice, the training module **660** may be part of the analysis model **614**.

[0067] The control system **612** includes, or stores, training set data **642**. The training set data **642** includes historical thin-film spectrum data **664** and contamination labels data **666**. The historical thin-film spectrum data **664** includes emission spectra for large number of thin-films. The contamination labels data **666** includes, for each emission spectrum in the historical thin-film spectrum data **664**, data indicating whether or not the emission spectrum corresponds to a contaminated thin-film. As will be set forth in more detail below, the training module **660** utilizes the historical thin-film spectrum data **664** and the contamination labels data **666** to train the analysis model **614** with a machine learning process.

[0068] In one embodiment, the historical thin-film spectrum data **664** includes data related to emissions spectra for a large number of thin-films. The emission spectra include the spectrum of photons or photoelectrons emitted by each of a large number of thin-films. In the case of photons spectra, each spectrum includes the distribution of photons of various energies, wavelengths, or frequencies for a particular previously analyzed thin-film. In the case of the electron spectra, each spectrum includes the distribution of energies of electrons for a particular previously analyzed thin-film.

[0069] In one embodiment, the contamination labels data **666** includes for each historical thin-film spectrum in the historical thin-film spectrum data **664**, a respective label. Each label indicates either contamination or non-contamination. Because there can be multiple types of contamination, there can be multiple types of contamination labels. In other words, labels indicating contamination can fall into one of a plurality of categories of contamination depending on the various ways that a thin-film can be contaminated during a thin-film deposition process.

[0070] In one embodiment the analysis model **614** includes a neural network. Training of the analysis model **614** will be described in relation to a neural network. However, other types of analysis models or algorithms can be used without departing from the scope of the present disclosure. The training module **660** utilizes the training set data **642** to train the neural network with a machine learning process. During the training process, the neural network receives, as input, historical thin-film spectrum data **664** from the training set data **662**. During the training process, the neural network outputs predicted category data. The predicted category data predicts, for each thin-film spectrum provided to the analysis model **614**, the category into which the spectrum falls. The categories can include contamination, non-contamination, or various individual categories of contamination and non-contamination. The training process trains the neural network to generate predicted category data that matches the contamination labels data **666** for each thin-film spectrum.

[0071] In one embodiment, the neural network includes a plurality of neural layers. The various neural layers include neurons that define one or more internal functions. The internal functions are based on weighting values associated with neurons of each neural layer of the neural network. During training, the control system **612** compares, for each set of historical thin-film spectrum data, the predicted category data to the actual label from the contamination labels data **666**. The control system generates an error function indicating how closely the predicted category data matches the contamination labels data **666**. The control system **612** then adjusts the internal functions of the neural network. Because the neural network generates predicted category data based on the internal functions, adjusting the internal functions will result in the generation of different predicted category data for a same set of historical thin-film spectrum data. Adjusting the internal functions can result in predicted category data that produces larger error functions (worse matching to the

contamination labels data **666**) or smaller error functions (better matching to the contamination labels data **666**).

[0072] After adjusting the internal functions of the neural network, the historical thin-film spectrum data **664** is again passed to the neural network and the analysis model **614** again generates predicted category data. The training module **660** again compares the predicted category data to the contamination labels data **666**. The training module **660** again adjusts the internal functions of the neural network. This process is repeated in a very large number of iterations of monitoring the error functions and adjusting the internal functions of the neural network until a set of internal functions is found that results in predicted category data that matches the contamination labels data **666** across the entire training set.

[0073] At the beginning of the training process, the predicted category data likely will not match the contamination labels data **666** very closely. However, as the training process proceeds through many iterations of adjusting the internal functions of the neural network, the errors functions will trend smaller and smaller until a set of internal functions is found that results in predicted category data that match the contamination labels data **666**. Identification of a set of internal functions that results in predicted category data that matches the contamination labels data **666** corresponds to completion of the training process. Once the training process is complete, the neural network is ready to be used to adjust thin-film deposition process parameters.

[0074] In one embodiment, after the analysis model **614** has been trained, the analysis model **614** can be utilized to analyze emission spectra of thin-films. In particular, as described in relation to FIGS. 1-5, a thin-film can be irradiated with an excitation light from a radiation source. The thin-film will then output emissions, such as photoluminescence photons or photoelectrons, as the case may be. A radiation sensor can then sense the energy spectrum of the emissions. The radiation sensor can pass the sensor signals to the analysis model **614**. The analysis model **614** analyzes the spectrum data and the signs the spectrum to a category. The category can include contamination, non-contamination, or various subcategories of contamination and non-contamination. The control system **612** can output an alert and stop further thin-film deposition processes if the category indicates contamination.

[0075] In one embodiment, the control system **612** includes processing resources **668**, memory resources **670**, and communication resources **672**. The processing resources **668** can include one or more controllers or processors. The processing resources **668** are configured to execute software instructions, process data, make thin-film deposition control decisions, perform signal processing, read data from memory, write data to memory, and to perform other processing operations. The processing resources **668** can include physical processing resources **668** located at a site or facility of the thin-film deposition system. The processing resources can include virtual processing resources **668** remote from the site thin-film deposition system or a facility at which the thin-film deposition system is located. The processing resources **668** can include cloud-based processing resources including processors and servers accessed via one or more cloud computing platforms.

[0076] In one embodiment, the memory resources **670** can include one or more computer readable memories. The memory resources **670** are configured to store software instructions associated with the function of the control system and its components, including, but not limited to, the analysis model **614**. The memory resources **670** can store data associated with the function of the control system **612** and its components. The data can include the training set data **642**, current process conditions data, and any other data associated with the operation of the control system **612** or any of its components. The memory resources **670** can include physical memory resources located at the site or facility of the thin-film deposition system **100**. The memory resources can include virtual memory resources located remotely from site or facility of the thin-film deposition system **100**. The memory resources **670** can include cloud-based memory resources accessed via one or more cloud computing platforms.

[0077] In one embodiment, the communication resources can include resources that enable the

control system **612** to communicate with equipment associated with the thin-film deposition system **100**. For example, the communication resources **672** can include wired and wireless communication resources that enable the control system **612** to receive the sensor data associated with the thin-film deposition system and to control equipment of the thin-film deposition system. The communication resources **672** can enable the control system **612** to control the various components of the thin-film deposition system. The communication resources **672** can enable the control system **612** to communicate with remote systems. The communication resources **672** can include, or can facilitate communication via, one or more networks such as wide networks, wireless networks, the Internet, or an intranet. The communication resources **672** can enable components of the control system **612** to communicate with each other.

[0078] In one embodiment, the analysis model **614** is implemented via the processing resources **668**, the memory resources **670**, and the communication resources **672**. The control system **612** can be a dispersed control system with components and resources and locations remote from each other and from the thin-film deposition system.

[0079] The components, functionality, and processes described in relation to the control system **612** and the analysis model **614** can be extended to the control systems and analysis models described in relation to FIGS. **1-5**.

[0080] FIG. **7** is a flow diagram of a process **700** for training an analysis model to determine whether a thin-film is contaminated or not, according to one embodiment. The process **700** can be implemented utilizing systems, components, and processes described in relation to FIGS. **1-6**. The process **700** can also be implemented using other systems, components, and processes. One example of an analysis model is the analysis model **614** of FIG. **6**, though the process **700** of FIG. **7** can apply to the analysis models described in any of the previous figures. The various steps of the process **700** can utilize components, processes, and techniques described in relation to FIGS. **1-6**. Accordingly, FIG. **7** is described with reference to FIGS. **1-6**.

[0081] At **702**, the process **700** gathers training set data including historical thin-film spectrum data and contamination labels data. One example of training set data is the training set data **662** of FIG. **6**. This can be accomplished by using a data mining system or process. The data mining system or process can gather training set data by accessing one or more databases associated with thin-film deposition systems and collecting and organizing various types of data contained in the one or more databases. The data mining system or process, or another system or process, can process and format the collected data in order to generate a training set data. The training set data can include historical thin-film spectrum data and contamination labels data as described in relation to FIG. **6**.

[0082] At **704**, the process **700** inputs historical thin-film spectrum data to the analysis model. In one example, this can include inputting contamination labels data to the analysis model with a training module as described in relation to FIG. **6**. One example of contamination labels data is the contamination labels data **666** of FIG. **6**. The historical thin-film spectrum data can be provided in consecutive discrete sets to the analysis model. One example of historical thin-film data is the historical thin-film data **664** of FIG. **6**. Each discrete set can correspond to a single thin-film or to groups of thin-film. The historical thin-film spectrum data can be provided as vectors to the analysis model. Each set can include one or more vectors formatted for reception and processing by the analysis model. The historical thin-film spectrum data can be provided to the analysis model in other formats without departing from the scope of the present disclosure.

[0083] At **706**, the process **700** generates predicted category data based on historical thin-film spectrum data. In particular, the analysis model generates, for each set of historical thin-film spectrum data, predicted category data. The predicted category data corresponds to a prediction category of contamination or non-contamination. This can be performed by the training module **660** or the analysis model **614** of FIG. **6**.

[0084] At **708**, the predicted category data is compared to the historical thin-film spectrum data **664**. In particular, the predicted category data for each set of historical thin-film spectrum data is

compared to the contamination labels data associated with that set of historical thin-film spectrum data. The comparison can result in an error function indicating how closely the predicted category data matches the contamination labels data. This comparison is performed for each set of predicted category data. In one embodiment, this process can include generating an aggregated error function or indication indicating how the totality of the predicted category data compares to the contamination labels data. These comparisons can be performed by the training module or by the analysis model. The comparisons can include other types of functions or data than those described above without departing from the scope of the present disclosure. This can be performed by the training module **660** or the analysis model **614** of FIG. **6**.

[0085] At **710**, the process **700** determines whether the predicted category data matches the contamination labels data based on the comparisons generated at step **708**. In one example, if the aggregate error function is less than an error tolerance, then the process **700** determines that the predicted category data does not match the contamination labels data. In one example, if the aggregate error function is greater than an error tolerance, then the process **700** determines that the predicted category data does match the contamination labels data. In one example, the error tolerance can include a tolerance between 0.1 and 0. In other words, if the aggregate percentage error is less than 0.1, or 10%, then the process **700** considers that the predicted category data matches the contamination labels data. If the aggregate percentage error is greater than 0.1 or 10%, then the process **700** considers that the predicted category data does not match the contamination labels data. Other tolerance ranges can be utilized without departing from the scope of the present disclosure. Error scores can be calculated in a variety of ways without departing from the scope of the present disclosure. The training module or the analysis model can make the determinations associated with process step **710**. This can be performed by the training module **660** or the analysis model **614** of FIG. **6**.

[0086] In one embodiment, if the predicted category data does not match the historical thin-film spectrum data at step **710**, then the process proceeds to step **712**. At step **712**, the process **700** adjusts the internal functions associated with the analysis model. In one example, the training module adjusts the internal functions associated with the analysis model. From step **712**, the process returns to step **704**. At step **704**, the historical thin-film spectrum data is again provided to the analysis model. Because the internal functions of the analysis model have been adjusted, the analysis model will generate different predicted category data than in the previous cycle. The process proceeds to steps **706**, **708** and **710** and the aggregate error is calculated. If the predicted category data does not match the contamination labels data, then the process returns to step **712** and the internal functions of the analysis model are adjusted again. This process proceeds in iterations until the analysis model generates predicted category data that matches the contamination labels data. This can be performed by the training module **660** or the analysis model **614** of FIG. **6**.

[0087] In one embodiment, if the predicted category data matches the historical contamination labels data and process step **710**, then the process **700** proceeds to **714**. At step **714** training is complete. The analysis model is now ready to be utilized to identify process conditions can be utilized in thin-film deposition processes performed by the thin-film deposition system. The process **700** can include other steps or arrangements of steps than shown and described herein without departing from the scope of the present disclosure.

[0088] FIG. **8** is a flow diagram of a thin-film deposition method **800**, according to one embodiment. The method **800** can be implemented utilizing systems, components, and processes described in relation to FIGS. **1-7**. The method **800** can also be implemented using other systems, components, and processes. At **802**, the method **800** includes depositing a thin-film on a patterned wafer in a thin-film deposition chamber. One example of a wafer is the wafer **106** of FIG. **1**. One example of a thin-film deposition chamber is the thin-film deposition chamber **102** of FIG. **1**. At **804**, the method **800** includes irradiating, in the thin-film deposition chamber, the thin-film with a laser. One example of a laser is the excitation laser **208** of FIG. **2**. At **806**, the method **800** includes

collecting photoluminescence spectrum data with a light sensor responsive to irradiation with the laser. One example of a light sensor is the light sensor **210** of FIG. **2**. At **808**, the method **800** includes analyzing the spectrum with an analysis model trained with a machine learning process. One example of an analysis model is the analysis model **214** of FIG. **2**. At **810**, the method **800** includes determining, with the analysis model, whether there is an oxygen leak based on the spectrum. At **812**, the method **800** includes, if there is an oxygen leak, outputting a signal indicating the oxygen leak.

[0089] FIG. **9** is a flow diagram of a thin-film deposition method **900**, according to one embodiment. The method **900** can be implemented utilizing systems, components, and processes described in relation to FIGS. **1-8**. The method **900** can also be implemented using other systems, components, and processes. At **902**, the method **900** includes depositing a thin-film on a patterned wafer in a first chamber. One example of a wafer is the wafer **306** of FIG. **3**. One example of a first chamber is the thin-film deposition chamber **302** of FIG. **3**. At **903**, the method **900** includes transferring the wafer from the first chamber to a second chamber. One example of a second chamber is the thin-film analysis chamber **334** of FIG. **3**. At **904**, the method **900** includes irradiating, in the second chamber, the thin-film with a laser. One example of a laser is the excitation laser **208** of FIG. **2**. At **906**, the method **900** includes collecting photoluminescence spectrum data with a light sensor responsive to irradiation with the laser. One example of a light sensor is the light sensor **210** of FIG. **2**. At **908**, the method **900** includes analyzing the spectrum with an analysis model trained with a machine learning process. One example of an analysis model is the analysis model **314** of FIG. **3**. At **910**, the method **900** includes determining, with the analysis model, whether there is an oxygen leak based on the spectrum. At **912**, the method **900** includes, if there is an oxygen leak, outputting a signal indicating the oxygen leak.

[0090] FIG. **10** is a flow diagram of a thin-film deposition method **1000**, according to one embodiment. The method **1000** can be implemented utilizing systems, components, and processes described in relation to FIGS. **1-9**. The method **1000** can also be implemented using other systems, components, and processes. At **1002**, the method **1000** includes depositing a thin-film on a patterned wafer in a thin-film deposition chamber. One example of a wafer is the wafer **106** of FIG. **1**. One example of a thin-film deposition chamber is the thin-film deposition chamber **102** of FIG. **1**. At **1004**, the method **1000** includes irradiating, in the thin-film deposition chamber, the thin-film with x-rays. At **1006**, the method **1000** includes collecting photoelectron spectrum data with a photoelectron sensor responsive to irradiation with the x-rays. One example of a photoelectron sensor is the emissions sensor **110** of FIG. **1**. At **1008**, the method **1000** includes analyzing the spectrum with an analysis model trained with a machine learning process. One example of an analysis model is the analysis model **114** of FIG. **1**. At **1010**, the method **1000** includes determining, with the analysis model, whether there is an oxygen leak based on the spectrum. At **1012**, the method **1000** includes, if there is an oxygen leak, outputting a signal indicating the oxygen leak.

[0091] FIG. **11** is a flow diagram of a thin-film deposition method **1100**, according to one embodiment. The method **1100** can be implemented utilizing systems, components, and processes described in relation to FIGS. **1-10**. The method **1100** can also be implemented using other systems, components, and processes. At **1102**, the method **1100** includes depositing a thin-film on a patterned wafer in a first chamber. One example of a wafer is the wafer **306** of FIG. **3**. One example of a first chamber is the thin-film deposition chamber **302** of FIG. **3**. At **1103**, the method **1100** includes transferring the wafer from the first chamber to a second chamber. One example of a second chamber is the thin-film analysis chamber **334** of FIG. **3**. At **1104**, the method **1100** includes irradiating, in the second chamber, the thin-film with x-rays. At **1106**, the method **1100** includes collecting photoelectron spectrum data with a photoelectron sensor responsive to irradiation with the x-rays. One example of a photoelectron sensor is the emissions sensor **310** of FIG. **3**. At **1108**, the method **1100** includes analyzing the spectrum with an analysis model trained with a machine

learning process. One example of an analysis model is the analysis model **314** of FIG. **3**. At **1110**, the method **1100** includes determining, with the analysis model, whether there is an oxygen leak based on the spectrum. At **1112**, the method **1100** includes, if there is an oxygen leak, outputting a signal indicating the oxygen leak.

[0092] FIG. **12** is a flow diagram of a thin-film deposition method **1200**, according to one embodiment. The method **1200** can be implemented utilizing systems, components, and processes described in relation to FIGS. **1-11**. The method **1200** can also be implemented using other systems, components, and processes. In one embodiment, at **1202**, the method **1200** includes depositing a thin-film on a wafer in a thin-film deposition chamber. One example of thin film deposition chamber is the thin film deposition chamber **102** of FIG. **1**. At **1204**, the method **1200** includes irradiating the thin-film with excitation light. At **1206**, the method **1200** includes detecting an emission spectrum from the thin-film responsive to the excitation light. At **1208**, the method **1200** includes detecting contamination of the thin-film by analyzing the emission spectrum with an analysis model of a control system. One example of an analysis model is the analysis model **614** of FIG. **6**. One example of a control system is the control system **612** of FIG. **6**.

[0093] FIG. **13** is a flow diagram of a thin-film deposition method **1300**, according to one embodiment. The method **1300** can be implemented utilizing systems, components, and processes described in relation to FIGS. **1-12**. The method **1300** can also be implemented using other systems, components, and processes. At **1302**, the method **1300** includes training, with a machine learning process utilizing data from a plurality of spectrums detected under a plurality of contamination and non-contamination conditions, an analysis model to detect contamination of thin-films. One example of an analysis model is the analysis model **614** of FIG. **6**. At **1304**, method **1300** includes depositing a thin-film on a wafer in a thin-film deposition chamber. At **1306**, the method **1300** includes irradiating the wafer with excitation light. One example of a thin-film deposition chamber is the thin-film deposition chamber **102** of FIG. **1**. At **1308**, the method **1300** includes detecting an emission spectrum from the wafer responsive to the excitation light. At **1310**, the method **1300** includes detecting whether or not the thin-film is contaminated by analyzing the emission spectrum with the analysis model.

[0094] In one embodiment, a method includes depositing a thin-film on a wafer in a thin-film deposition chamber and irradiating the thin-film with excitation light. The method includes detecting an emission spectrum from the thin-film responsive to the excitation light and detecting contamination of the thin-film by analyzing the emission spectrum with an analysis model of a control system.

[0095] In one embodiment, a system includes a thin-film deposition chamber configured to deposit a thin-film on a wafer and a radiation source configured to irradiate the thin-film with excitation light. The system includes an emissions sensor configured to detect an emission spectrum from the wafer responsive to the excitation light. The system includes a control system coupled to the radiation source and the emissions sensor and configured to detect contamination of the thin-film by analyzing the spectrum of photons and to stop thin-film deposition processes in the thin-film deposition chamber responsive to detecting contamination of the thin-film.

[0096] In one embodiment, a method includes training, with a machine learning process utilizing data from a plurality of spectrums detected under a plurality of contamination and non-contamination conditions, an analysis model to detect contamination of thin-films. The method includes depositing a thin-film on a wafer in a thin-film deposition chamber and irradiating the wafer with excitation light. The method includes detecting an emission spectrum from the wafer responsive to the excitation light and detecting whether or not the thin-film is contaminated by analyzing the emission spectrum with the analysis model.

[0097] Embodiments of the present disclosure are able to detect contamination of thin-films in situ. Thin film deposition processes can be stopped upon detection of contamination, thereby reducing the number of wafers with faulty thin-films. This provides many benefits including improved wafer

yield and fewer scrapped wafers. Furthermore, oxygen leaks in deposition chambers can be detected based on the detected contamination.

[0098] The various embodiments described above can be combined to provide further embodiments. All U.S. patent application publications and U.S. patent applications referred to in this specification and/or listed in the Application Data Sheet are incorporated herein by reference, in their entirety. Aspects of the embodiments can be modified, if necessary, to employ concepts of the various patents, applications and publications to provide yet further embodiments.

[0099] These and other changes can be made to the embodiments in light of the above-detailed description. In general, in the following claims, the terms used should not be construed to limit the claims to the specific embodiments disclosed in the specification and the claims, but should be construed to include all possible embodiments along with the full scope of equivalents to which such claims are entitled. Accordingly, the claims are not limited by the disclosure.

Claims

1. A thin-film deposition method, comprising: depositing a thin-film on a wafer in a first chamber; transferring the wafer to a second chamber communicatively coupled to the first chamber; irradiating the thin-film with excitation light with a radiation source in the second chamber; sensing, with an emissions sensor, an emission spectrum from the thin-film responsive to the excitation light in the second chamber; and detecting contamination of the thin-film by analyzing the emission spectrum with an analysis model of a control system.
2. The method of claim 1, further comprising maintaining a vacuum condition of the deposition chamber in the detection chamber while irradiating the thin-film with excitation light.
3. The method of claim 1, wherein detecting contamination includes detecting oxidation of the thin-film.
4. The method of claim 3, further comprising detecting, with the control system, a leak in the first chamber based on detecting oxidation of the thin-film.
5. The method of claim 1, further comprising halting operation of the first chamber responsive to detecting contamination of the thin-film.
6. The method of claim 1, wherein the excitation light includes ultraviolet light, wherein detecting the emission spectrum includes detecting a photoluminescence spectrum.
7. The method of claim 1, wherein the excitation light includes X-ray light, wherein detecting the emission spectrum includes detecting a photoelectron spectrum.
8. The method of claim 1, wherein the excitation light is laser light.
9. The method of claim 1, wherein the radiation source and the emission sensor are positioned in a tube.
10. The method of claim 1, further comprising training the analysis model with a machine learning process to detection contamination of the thin-film.
11. A system, comprising: a first chamber configured to deposit a thin-film on a wafer; a second chamber communicatively coupled to the first chamber and including: a radiation source configured to irradiate the thin-film with excitation light; and an emissions sensor configured to detect an emission spectrum from the wafer responsive to the excitation light; and a control system coupled to the radiation source and the emissions sensor and configured to detect contamination of the thin-film by analyzing the emission spectrum and to stop thin-film deposition processes in the first chamber responsive to detecting contamination of the thin-film.
12. The system of claim 11, wherein the radiation source is a laser, wherein the emissions sensor is an ultraviolet light sensor.
13. The system of claim 11, wherein the radiation source is an x-ray source, wherein the emissions sensor is a photoelectron spectroscopy detector.
14. The system of claim 11, wherein the radiation source is positioned to irradiate the thin-film in

the second deposition chamber.

15. The system of claim 11, comprising a tube including the radiation source and the emissions sensor.

16. The system of claim 11, wherein the control system includes an analysis model trained with a machine learning process to detect contamination of thin-films based on the emission spectrum.

17. The system of claim 16, wherein the analysis model includes a neural network.

18. A method, comprising: training, with a machine learning process utilizing data from a plurality of spectrums detected under a plurality of contamination and non-contamination conditions, an analysis model to detect contamination of thin-films; depositing a thin-film on a wafer in a first deposition chamber); transferring the wafer to a second chamber communicatively coupled to the first chamber; irradiating, with a radiation source in the second chamber, the wafer with excitation light; detecting, with an emissions sensor in the second chamber, an emission spectrum from the wafer responsive to the excitation light; and detecting whether or not the thin-film is contaminated by analyzing the emission spectrum with the analysis model.

19. The method of claim 18, wherein the thin-film includes titanium nitride, wherein detecting contamination of the titanium nitride includes detecting that the emission spectrum includes emissions characteristic of titanium oxide.

20. The method of claim 19, further comprising depositing the thin-film over a semiconductor nanosheet structure.
